

SPELLZEE ERP

Product & Business Requirements

Draft 2

Student Operations, Academic Delivery, Teacher & HR, Finance, Customer Success, Governance & Capacity Planning

Status: Working Draft — Product Discovery
Current Build Focus: Operations / Delivery Platform
Long-Term Direction: Integrated Spellzee ERP

This document consolidates the business problems, product direction, workflows and controls discussed during discovery. It is a product/business requirements baseline, not yet a technical architecture or final database specification.

1. Executive Summary

Spellzee is moving from a collection of operational tools toward a controlled, integrated ERP. The immediate implementation focus is Operations / Delivery, while the product foundation must support Academic, Teacher & HR, Finance, Customer Success, Sales, Communication, Governance, Analytics and future AI.

Core principle: The ERP should protect the student relationship, preserve history, enforce business rules, make capacity visible, establish clear ownership, and ensure important actions are traceable and appropriately approved.

2. Product Vision

Spellzee ERP should become the operational system of record for the student's complete relationship with Spellzee.

- One permanent student identity across admissions, enrollments, subscriptions, payments, classes, teachers, attendance, lessons, assessments, communication, support and retention.
- One connected teacher workforce view covering availability, allocation, training, quality, incentives and payroll.
- One controlled workflow for scheduling, rescheduling, compensation, breaks and teacher/student changes.
- One governance layer for permissions, approvals, segregation of duties, duplicate prevention and audit.
- One capacity view showing current and forecast delivery capability.
- AI should be layered onto reliable data and workflows after the foundation is proven.

3. Proposed ERP Domains

Domain	Scope
Identity & Master Data	Parent, Student, Employee, Teacher, Course, Subject and permanent identifiers.
Sales & Admissions	Lead/demo/admission context; existing sales systems can remain integrated during the initial Operations build.
Student & Customer	Student 360, parent relationships, enrollment, lifecycle and customer history.
Operations / Delivery	Handover, coordinator ownership, allocation, scheduling, compensation, SLA and tickets.
Academic	Curriculum, lessons, worksheets, activities, assessments, attendance, progress and recordings.
Teacher & HR	Recruitment, profile, availability, training, certification, observation, performance, incentives, leave and payroll.
Finance	Subscriptions, payments, credits, refunds, controls and compensation.
Communication	WhatsApp, in-app communication, email, calls and notifications.
Governance & Administration	Roles, permissions, approvals, segregation of duties, audit, duplicate prevention and security.
Analytics & Management	Operations, academic, teacher, finance, capacity and retention dashboards.
AI & Automation	Future intelligent matching, summaries, risk prediction, forecasting and workflow automation.

4. Student Identity & Data Integrity

4.1 Permanent Student Identity

Every student receives a Spellzee-generated permanent ID, for example STU-2026-000184. It never changes when the student changes course, teacher, schedule, subscription or contact details, takes a break, or returns later.

4.2 Separate Business Objects

Recommended chain: Student → Enrollment → Subscription → Payment; Enrollment → Class Schedule → Individual Sessions. Teacher assignments, attendance, lessons, assessments, tickets and communications reference the correct underlying record.

4.3 Duplicate Prevention

- Before creating a student, search for likely existing records using approved matching signals such as names, parent details, phone numbers, alternate numbers and email addresses.
- Show possible matches before creating a new identity.
- High-confidence duplicate creation should be blocked or require authorized approval.
- Keep historical contact details without changing the permanent identity.
- Support controlled merging of duplicate records while preserving history and audit references.

4.4 Controlled Deletion & Audit

Critical records should normally be archived/cancelled rather than permanently deleted. Important changes must record user, timestamp, old value, new value, reason and approval where required.

5. Admission Handover & Onboarding

- Capture parent/student details, course/subject, preferred days/times and operational requirements after admission.
- Carry demo assessment and teacher feedback into Operations.
- Verify information with the parent and record the outcome.
- Assign coordinator ownership and start the applicable SLA.
- Match teacher/schedule using subject, level, skills, language and capacity.
- Confirm allocation, create/update the live-class schedule through the LMS integration and notify parent/teacher.
- Give the teacher preparation context, including relevant demo feedback.

6. Coordinator & Ownership Management

Coordinator responsibility should be an explicit data object, not an assumption based on the last person who contacted the parent.

- Current coordinator/owner and assigned students/parents.
- Responsibility start/end dates.
- Lifecycle role: Onboarding Owner, Student Success Owner, Retention Owner, Operations Owner, Academic Owner, Ticket Owner, etc.
- Ownership-change history and escalation owner.

The current business model can be represented as a lifecycle: onboarding owner until first class → designated coordinator during the first month → later Student Success/Retention ownership, with exact policy configurable.

7. Class Allocation, Scheduling & Reallocation

- Allocation means teacher assignment plus schedule/arrangement.
- Support new allocation, teacher change, schedule/day change, course/subject change, session-type change, break/resume and other reallocation.
- Preserve previous allocations; record reason, initiator, effective date and approval.
- Separate recurring schedules from individual sessions.

- Map Spellzee IDs to external LMS IDs and keep Spellzee as the core operational source of truth.

8. Session, Attendance, Cancellation & Compensation

Principle: A purchased session should never be silently lost, and a missed session should never be silently forgiven. Every session outcome must have a reason, policy outcome, credit impact and history.

8.1 Session Outcomes

- Completed / attended
- Student absent
- Advance parent cancellation
- Late cancellation
- Teacher absent
- Teacher-side technical issue
- Student-side technical issue
- Spellzee cancellation
- No-show
- Rescheduled
- Compensation required
- Compensation completed

8.2 Reminder & Cancellation Rules

For a 3 PM class, the system can send a configurable reminder several hours beforehand and optionally a closer reminder. A configurable cancellation cutoff determines whether the session credit is protected, consumed or handled by another approved rule.

8.3 Compensation Rules

Teacher absence or verified teacher/Spellzee-side failure should normally protect the student's entitlement and trigger compensation according to policy. Compensation must be a separate session and must not alter the normal recurring schedule.

8.4 Rescheduling Controls

Management should configure notice requirements, rescheduling limits, compensation validity, repeated-absence thresholds and escalation rules so the system does not allow endless rescheduling.

8.5 Session Ledger

Maintain purchased, completed, protected, consumed, compensated and remaining session counts, each linked to individual sessions and reasons.

9. Completion & Payment Forecast

For session-based subscriptions, calculate a Planned Completion Date from start date, session entitlement and recurring schedule. Calculate a Projected Completion Date from actual attendance, cancellations, compensation and schedule changes.

- Planned completion date
- Projected completion date
- Delay versus plan
- Missed sessions by reason

- Compensation backlog
- Internal expected renewal/payment timing

Internal forecasts can support management and Finance without necessarily exposing all calculations to parents.

10. Parent Portal & Communication

- My Children and current status.
- Current coordinator and appropriate contact details.
- Upcoming/completed classes and class access.
- Attendance, missed sessions and compensation.
- Worksheets, materials, activities and permitted assessments.
- Teacher information and permitted recordings.
- Academic progress and feedback.
- Subscription/session balance and appropriate payment information.
- Messages and Help/Support.
- Ticket tracking where a conversation becomes an operational issue.

Communication and ticketing remain distinct: communication is the channel; a ticket is the tracked issue/request with owner, SLA and resolution.

11. Academic Management

- Enrollment, academic program, course, subject, level and curriculum.
- Lesson records: topics, activities, homework, outcomes and teacher notes.
- Student assessments and academic feedback.
- Worksheets/materials and student activities.
- Attendance based on reliable participation data.
- Progress tracking.
- Session recordings where permitted.
- Academic history available to future assigned teachers.

12. Student Continuity & Teacher Handover

When a teacher changes, the next teacher should receive a structured history instead of starting without context.

- Previous teachers and assignment history.
- Subjects/levels and topics already taught.
- Student strengths and difficulties.
- Demo and later assessment results.
- Relevant academic feedback.
- Attendance patterns.
- Important parent/student preferences.
- Open issues and relevant context.

Future AI: Generate a permission-controlled student history/learning summary from verified system records. The AI summary must not replace the underlying source records.

13. Teacher & HR Management

13.1 Teacher Master Profile

- Teacher ID and personal/contact details
- Qualifications and documents
- Subjects, levels and skills
- Languages/specializations
- Employment status
- Joining/exit information
- Payroll/bank information

13.2 Recruitment & Onboarding

Recruitment → screening → selection → joining → teacher profile → training → mock/assessment → certification/approval → eligible for allocation.

13.3 Teacher Training

The existing two-week onboarding program should be represented as a structured training program with modules, attendance/progress, mock classes, trainer assessment, pass/fail and certification.

13.4 Observation & Quality

Existing observation/audit rounds should capture teacher, session, observer/trainer, criteria, scores, feedback, action items, re-observation and outcome.

13.5 Incentives

Observation results can feed configurable incentive rules. For example, meeting defined criteria across an audit round can make a teacher eligible for a specified incentive percentage. The percentage and criteria should be configurable.

13.6 Leave & Payroll

Leave/unavailability feeds capacity planning. Approved work, incentives and other compensation inputs can feed payroll.

14. Capacity Planning & Demand Forecasting

Availability asks: Is a teacher free? **Capacity planning asks:** How many students can Spellzee realistically onboard, by when, and in which categories?

- Total and weekly teacher capacity.
- Occupied/allocated capacity.
- Current free slots.
- Scheduled capacity releases.
- Temporary capacity released by student breaks.
- Teacher leave/unavailability.
- Capacity by day/time slot.
- Capacity by subject, level, language and session type.
- Teacher utilization.
- Current and expected demand.
- Capacity gap/surplus.
- Recruitment requirement forecast.

14.1 Capacity vs Demand

Example: if current free capacity is 32 sessions, expected releases add 8 and expected demand is 45, the system can flag a projected gap of 5. Operations can then avoid making unsupported promises and HR can plan recruitment.

14.2 Skill & Language Capacity

Capacity should be filterable for requirements such as English, Tamil-English bilingual support, Hindi-English support, level, subject, 1-to-1/group and time windows.

14.3 Utilization

Show underused, balanced and overloaded teachers. Management should set a target utilization range rather than simply maximizing utilization.

14.4 Workforce Planning

Later, historical admissions, churn, breaks, teacher attrition, seasonality and skill demand can support predictive workforce planning.

15. Tickets, SLA & Retention

- Any new or existing student issue/request can create a ticket.
- Ticket stores student, parent, category, priority, team/person, SLA, status, communication, resolution and history.
- Categories include class, teacher, academic, schedule, subscription and general.
- Workflow: Open → In Progress → Resolved, with escalation.
- Current discussed target: 48-hour ticket resolution, subject to final policy definition.
- Retention calls can trigger after a configurable session count such as every three sessions.
- Student health can combine attendance, rescheduling, complaints, progress, breaks, teacher changes and subscription status.

16. Finance & Subscription

- Upfront subscription/payment records.
- Permanent mapping of payments to the appropriate student/subscription.
- Session entitlement and consumption.
- Remaining sessions and validity.
- Refund/credit controls.
- Renewal/payment forecast.
- Teacher incentive/payroll inputs.
- Stronger approval and audit for financial changes.

17. Governance, Segregation of Duties & Approvals

17.1 Role-Based Access

Permissions should determine who can view, create, edit, approve, cancel, merge, archive, export or administer information.

17.2 Maker–Checker

For sensitive transactions, the requester should not approve their own request. Examples include refunds, duplicate merges, historical corrections, high-impact subscription changes and restricted deletions.

17.3 Approval Hierarchy

A future permission matrix can distinguish staff, team lead, manager, department head and restricted administrator authority. Exact roles and thresholds remain management decisions.

17.4 Overrides

Authorized overrides must record who, what, why, old value, new value and approval.

17.5 Data Integrity

- Unique IDs for major entities.
- Valid relationships between student, enrollment, subscription, payment, class and session.
- No orphan financial/class records.
- Controlled state transitions.
- Duplicate detection and controlled merging.
- Complete auditability of sensitive changes.

18. Integrations

Current ecosystem includes Merithub for live-class/LMS infrastructure, TeliCRM for student/sales data, FreeJump for calling activity, WhatsApp for communication and spreadsheets for allocation. The future architecture should use a Spellzee Integration Layer.

- Spellzee identifiers remain the core identity.
- Store external IDs as mappings.
- Map LMS user/class/session IDs to Spellzee records.
- Make API failures visible and retryable.
- Validate and log webhook events.
- External systems should not silently overwrite core Spellzee records.

19. Analytics & Management

- Admissions and allocation pipeline.
- SLA performance.
- Today's/upcoming classes.
- Teacher utilization and capacity.
- Capacity vs demand and recruitment forecast.
- Teacher observation/performance.
- Attendance and completion delay.
- Compensation backlog.
- Ticket/SLA performance.
- Student health and retention risk.
- Subscription consumption and renewal forecast.
- Reallocation/cancellation reasons.

20. AI & Automation — Later

- Teacher/student allocation recommendations.

- Duplicate detection assistance.
- Student history summaries for new teachers.
- Retention-risk prediction and recommended actions.
- Academic progress summaries.
- Teacher observation summaries and coaching recommendations.
- AI analysis of employee sales pitches/calls.
- Demand and workforce forecasting.
- Workflow automation.

AI-generated outputs should respect permissions, be traceable to source data where appropriate, and never silently modify critical records.

21. Non-Functional Requirements

- Security and role-based access.
- Audit logs for critical actions.
- Reliable backup and recovery.
- Monitoring and error logging.
- Scalability for growing students, teachers and sessions.
- API-first integration capability.
- Development, test/staging and production environments.
- Documentation for architecture, database, APIs, deployment and operations.
- Company-controlled repositories, cloud accounts and secrets.
- No single-person dependency for critical system knowledge.

22. Suggested Implementation Roadmap

- Phase 0 — Discovery & Foundation: workflows, rules, roles, permissions, identifiers, current-system mapping.
- Phase 1 — Operations MVP: student identity, handover, coordinator ownership, teacher availability/capacity basics, allocation, scheduling, session ledger, LMS integration, notifications, tickets and audit.
- Phase 2 — Academic & Customer Success: lessons, assessments, worksheets/materials, progress, recordings, parent portal, retention.
- Phase 3 — Teacher & HR: recruitment, onboarding/training, certification, observations, performance, incentives, leave and payroll integration.
- Phase 4 — Finance & Enterprise Controls: subscriptions/payments, refunds, approvals and maker-checker.
- Phase 5 — Analytics & AI: management dashboards, capacity forecasting, predictive retention, intelligent matching and AI summaries.

23. Open Decisions for Next Discovery

- Exact SLA targets and definitions.
- Cancellation cutoff and late-cancellation rules.
- Reschedule/compensation limits and validity.
- Treatment of student-side versus teacher-side technical failures.
- Completion and renewal rules when sessions are delayed.

- Parent-facing versus internal forecast visibility.
- Approval matrix and financial thresholds.
- Fields that require approval to change.
- Duplicate matching confidence and merge process.
- Teacher utilization target and capacity calculation method.
- Academic curriculum/assessment model.
- Teacher payroll and incentive rules.
- Migration/ownership plan for current systems.
- WhatsApp provider/API and in-app communication scope.
- Privacy, data retention and access policies.

24. Glossary

Term	Meaning
Student	Permanent person identity.
Enrollment	Student's participation in a specific course/program.
Subscription	Commercial agreement or purchased session entitlement.
Payment	Financial transaction linked to the relevant commercial record.
Class Schedule	Recurring/planned arrangement for delivery.
Session	One specific class occurrence.
Allocation	Teacher assignment plus delivery schedule/arrangement.
Compensation Class	Separate session used to fulfil a protected/missed entitlement.
Capacity	Realistic delivery capability.
Utilization	Proportion of available capacity allocated/used.
SLA	Defined time target for completing an action.
Maker–Checker	One user requests/creates; another authorized user approves.
Audit Trail	History of important actions and changes.
RBAC	Role-Based Access Control.
Student 360	Connected operational, academic, commercial and support view.

25. Final Product Direction

Spellzee should not become a larger spreadsheet. It should become a controlled operating system for the business: one identity, connected workflows, preserved history, capacity visibility, clear ownership, governed changes, measurable delivery and—later—intelligent automation.

Draft 2 intentionally remains a product/business baseline. Technical architecture, database schema, API contracts, detailed UX and implementation tickets should follow after the business rules and workflows are sufficiently finalized.